CLASS 7 — SCIENCE · FORMULA SHEET

Chapter 4: **Changes Around Us: Physical and Chemical** | Quick revision, one page

Physical change

- No new substance is formed; only shape, size, state or appearance changes.
- Usually reversible, e.g. melting of ice, boiling of water, dissolving sugar, cutting paper, crystallisation.
- Properties such as colour and composition of the substance remain the same.

Chemical change

- One or more new substances with different properties are formed; usually irreversible.
- Examples: burning of paper or wood, rusting of iron, cooking food, souring of milk, ripening of fruit.
- Signs: change in colour, gas evolved, heat or light given out, sound, change in smell.

Rusting

- Iron + oxygen + water $\rightarrow$ rust (hydrated iron oxide, a brown flaky substance).
- Needs both air (oxygen) and moisture; salty water speeds it up.
- Prevention: painting, oiling or greasing, galvanisation (coating with zinc), using stainless steel.

Crystallisation and common tests

- Crystallisation: formation of crystals from a hot saturated solution on slow cooling (physical change).
- Baking soda + vinegar $\rightarrow$ carbon dioxide gas, which turns lime water milky.
- Burning of magnesium ribbon gives a dazzling white light and white ash (chemical change).

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